Little Explorers · Class 32

Countdown Rocket Launch

10, 9, 8… press the button… BLAST OFF!

Count down | Button | Blast off



Did you know?
Real rocket launches have a computer that counts down and checks everything is ready!

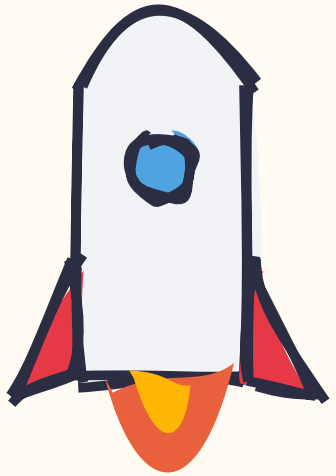
LEARN

Let's wonder...

Why do rockets count down?

10... 9... 8...

Before every real rocket launch, everyone counts down to zero. Why not just go? Because the computer is **checking** everything is ready, one step at a time!



For fun?

To get ready?

To be safe!

Today: you program a **countdown** and launch a rocket on the screen with a **real button**. Our last class: let's celebrate!

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Learning objectives

Today I can...

1	**Count down:** make the computer **count down** from 10 to 0 using a variable	**Proof:** my sprite counted 10, 9, 8... 0
2	**Real button:** press a **real button** to start the launch on the screen	**Proof:** my button started the countdown
3	**Blast off!:** make the rocket **fly up** and celebrate the end of our course	**Proof:** my rocket blasted off

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Objective 3 · Blast off! · video

Video time!



How Do Rockets Fly?

SciShow Kids · about 5 minutes · youtu.be/Lti6a_YYQl0

Watch for...

1. What pushes a rocket **up**?
2. Why do rockets need so much fuel?
3. Where do rockets go?

Scan to watch

After the video: Count down together from 10 to 0 and BLAST OFF with your arms!

Objective 1 · Count down

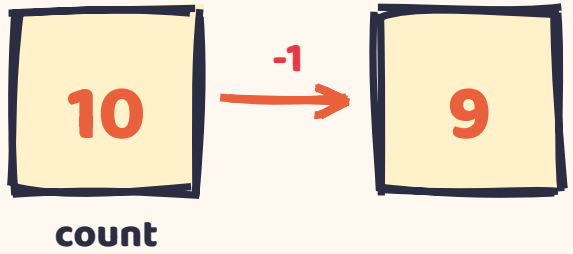
A variable is a box that holds a number

The idea

We make a box called **count**. Put **10** inside.
Then again and again: **say** the number, wait 1 second, take **1 away**.

10, 9, 8… when the box holds **0**: blast off!

Example: a sweet jar



A jar has 10 sweets. Each day you eat 1. The jar tells you how many are left: 9, 8, 7… A variable is a jar for numbers!

Objective 2 · Real button

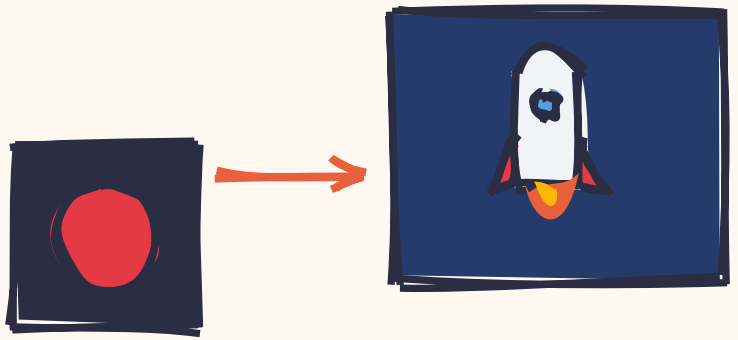
Real button, screen rocket

The idea

In **live mode**, mBlock talks to the board while the program runs.

Press the **real button** on the desk → the board tells the computer → the **rocket on the screen** starts the countdown!

Example: a TV remote



You press a button on the remote in your hand, and something changes on the TV screen. Our button is a remote for the rocket!

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Real world

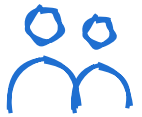
Countdowns everywhere



Rocket launch
Computers check everything before lift-off



Microwave
Counts down the seconds



Crossing light
Shows seconds left to cross



Races
Ready, steady, GO!

? Where have you seen numbers counting down?

Every class!
This year we used buttons, lights, sensors and code. Today we put them together!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human rocket launch
2. What we use
3. Snap the picture blocks
4. Make the countdown
5. Blast off!
6. Launch with the real button
7. Mission control challenges

Class game

Human rocket launch

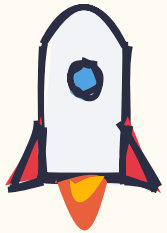
How to play

1. Everyone crouches down low: you are rockets!
2. The teacher holds up fingers: **10**. Count together, one each second.
3. At **0**: jump up and BLAST OFF with your arms!
4. Try counting from 5, then from 20. What changes?

Then discuss

What number did we start from?

What did we take away each time?



Big idea

A countdown starts at a number and takes 1 away each time.

Build it

What we use

Board + button
pre-wired by the teacher

mBlock 5
live mode, Rocketship sprite

LED
lights up at blast-off

Stickers
to decorate your launch pad

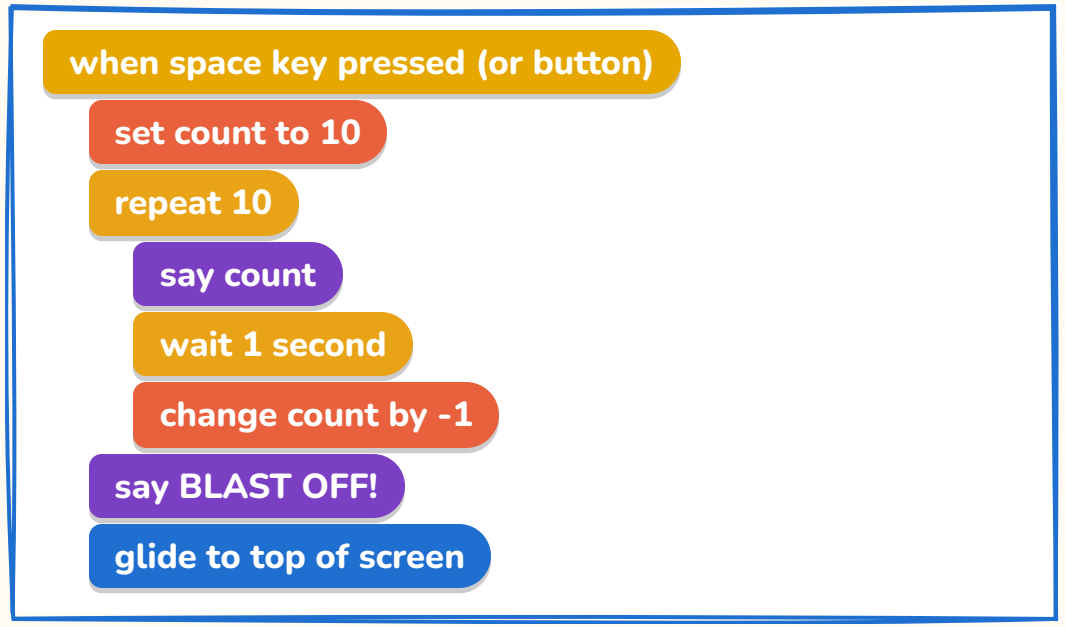
Plug in the USB cable. The teacher connects mBlock in live mode.

Careful
Keep water and snacks away from the laptop and board.

MAKE

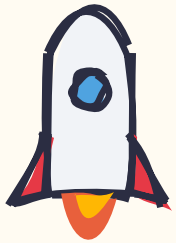
Time to build!

Snap the picture blocks



Read it like a story

Fill the box with 10.
Say it, wait, take 1 away.
Do it **10 times**.
Then: **BLAST OFF!** Fly up!



Make a variable
Variables → Make a Variable → call it 'count'.

MAKE

Objective 1 · Build step 1 of 3

Make the countdown

a Choose the **Rocketship** sprite. Pick a space backdrop.

b Make a variable called **count**.

c Snap the blocks: set count to 10, repeat 10: say, wait, change by -1.

d Press the space key to test.

e Does it count 10, 9, 8… 1?



Check: Your rocket counts down from 10.

MAKE

Objective 3 · Build step 2 of 3

Blast off!

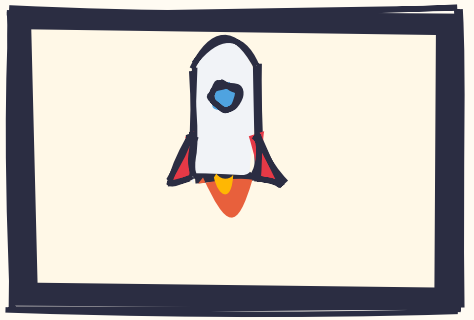
a After the repeat, add **say BLAST OFF!**

b Add **glide 2 secs to x: 0 y: 180**: the rocket flies up!

c Add a **sound**: a whoosh or a cheer.

d At the start, put the rocket back at the bottom.

e Test again!



Check: The rocket flies up after 0.

MAKE

Objective 2 · Build step 3 of 3

Launch with the real button

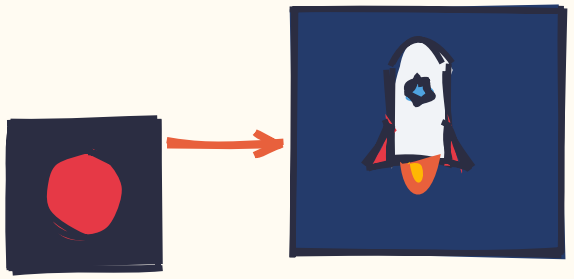
a Change the start block to **when button pressed** (the teacher shows you).

b Press the real button on your desk.

c Does the countdown start on the screen?

d Add: **turn LED on** at blast-off.

e Launch for your friends!



Check: The real button launches the rocket.

Press the button gently, don't bang it.

MAKE

Level up!

Mission control challenges

Choose your challenge



Count from 20: what do you need to change?



Sounds: play a beep for every number.



Two rockets: launch a race between two rockets!



Space trip: after blast-off, land on a new planet backdrop.

Think like an engineer

Why did we need to change TWO numbers to count from 20?

Boss level

Make the rocket shake before it takes off!

Objective check

Can I really do it? Prove it.

01	Show your countdown from 10 to 0.	○ ○ ○
02	Press the real button. What happens?	○ ○ ○
03	Tell me: what does the 'count' box do?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Launch party: everyone counts down together for each rocket. Celebrate the end of the course!

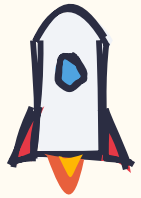
Show what you know

What did we learn?

- [x] A variable is a box that holds a number
- [x] Change by -1 makes it count down
- [x] A real button can control the screen
- [x] We built it all ourselves, well done, inventors!

At home

Count down before you do something fun at home: 5, 4, 3, 2, 1, GO!



Next class: Show your family what you made!

Exit ticket: what number is in the count box after 3 seconds?